

LAC EON, QC, Canada

WMO: 714210

Lat: 51.867N

Lon: 63.283W

Elev: 1932

StdP: 13.70

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-23.8	-35.8	0.8	-28.2	-31.3	1.0	-23.3	21.3	10.7	19.7	8.7	5.1	320	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	78.1	62.7	73.9	60.1	70.4	58.5	66.0	73.4	63.0	69.8	60.7	67.2	8.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	94.0	69.1	60.5	84.4	65.8	58.2	77.8	63.5	31.9	72.9	29.6	69.5	27.8	67.3	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.4	14.5	-36.9	84.6	4.7	5.6	-40.2	88.6	-43.0	91.8	-45.6	95.0	-49.0	99.0		
(5)			-37.4	68.4	4.8	3.6	-40.9	71.0	-43.7	73.1	-46.4	75.1	-49.8	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	-2.1	0.2	11.5	26.1	39.0	51.1	58.9	57.3	47.1	35.0	21.4	6.5
	DBStd	23.14	13.07	12.41	12.35	9.18	7.17	8.11	6.49	6.34	6.99	7.39	9.98	12.20
	HDD50	8179	1615	1395	1193	718	349	79	6	10	138	467	860	1349
	HDD65	13000	2080	1815	1658	1168	805	423	210	251	538	929	1309	1814
	CDD50	694	0	0	0	0	9	113	282	237	50	3	0	0
	CDD65	41	0	0	0	0	0	7	20	14	0	0	0	0
	CDH74	367	0	0	0	0	3	77	183	102	2	0	0	0
	CDH80	82	0	0	0	0	0	18	49	15	0	0	0	0
	WSAvg	7.1	7.0	7.2	7.9	7.8	6.9	7.0	6.6	6.4	7.2	7.3	7.3	7.0
	PrecAvg	41.7	2.8	2.5	3.2	2.4	3.1	3.4	4.4	4.6	3.6	4.3	3.8	3.6
(15) Precipitation	PrecMax	50.0	7.0	4.7	7.2	3.7	6.6	5.2	8.0	7.8	6.1	10.1	5.9	6.6
	PrecMin	31.1	0.8	0.4	0.9	0.4	1.3	1.9	2.0	1.4	1.9	1.8	1.7	0.7
	PrecStd	5.8	1.5	1.1	1.6	1.0	1.5	1.0	1.6	1.5	1.1	2.0	1.2	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	35.9	44.1	55.6	71.6	81.8	85.4	82.0	71.5	61.2	53.0	35.3
		MCWB	32.6	33.9	36.2	42.5	54.3	61.8	70.9	67.1	58.7	52.8	50.7	34.6
	2%	DB	29.4	31.5	37.0	47.9	64.7	74.7	78.6	76.7	66.6	54.1	43.1	32.2
		MCWB	28.9	30.2	32.0	37.3	49.2	58.5	62.9	63.7	56.7	48.8	41.3	31.8
	5%	DB	24.9	26.8	33.1	43.4	58.5	69.8	74.4	72.7	62.6	50.3	38.3	29.9
		MCWB	24.1	25.4	29.9	35.7	46.1	55.7	60.4	61.0	55.5	46.0	36.7	29.1
	10%	DB	19.0	20.8	30.1	39.5	53.5	65.6	70.7	68.7	58.8	46.5	34.3	25.2
		MCWB	17.9	19.1	26.9	33.7	43.9	54.0	59.1	59.1	53.4	43.2	32.8	24.2
	0.4%	WB	32.1	33.6	38.0	44.0	55.6	66.1	73.8	72.7	63.2	53.9	52.3	35.0
		MCDB	32.6	34.2	43.0	50.6	67.2	75.7	85.6	77.6	67.1	57.6	52.7	35.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.7	29.2	33.2	39.7	51.2	61.8	66.5	65.7	59.0	49.7	42.2	31.9
		MCDB	29.1	30.3	35.9	45.8	62.1	69.4	73.8	72.6	62.1	52.6	43.1	32.2
	5%	WB	23.9	24.5	30.3	36.8	47.8	58.9	63.4	62.7	56.1	46.9	37.3	29.2
		MCDB	24.6	26.0	33.3	41.8	56.7	67.2	70.7	69.1	61.1	49.6	38.5	29.9
	10%	WB	17.9	18.7	27.0	34.3	44.8	55.9	61.0	60.3	53.3	43.3	33.1	24.1
		MCDB	18.8	20.2	30.0	38.8	53.0	63.1	67.8	66.2	58.2	45.8	34.2	25.0
	5% DB	MDBR	18.8	20.8	22.1	18.8	18.4	18.7	17.4	16.5	14.6	11.3	12.5	15.7
		MCDBR	19.2	20.0	22.8	25.2	30.8	26.9	23.6	22.9	20.6	18.2	14.9	14.2
		MCWBR	17.9	18.9	19.2	17.0	17.5	12.8	10.8	11.9	12.8	12.7	14.2	14.6
		MCWBR	18.2	19.4	19.7	20.9	26.9	22.4	19.5	19.8	17.7	15.3	14.8	14.3
(35) Mean Daily Temperature Range	5% WB	MCWBR	18.3	18.0	16.9	15.0	16.6	12.6	11.0	12.5	13.1	12.5	14.7	14.5
	taub		0.235	0.245	0.263	0.318	0.350	0.353	0.387	0.352	0.328	0.294	0.279	0.241
		taud	2.303	2.293	2.274	2.190	2.373	2.432	2.366	2.471	2.508	2.571	2.400	2.288
(42) Clear-Sky Solar Irradiance	Ebn at Noon		254	284	299	291	286	286	273	279	274	265	233	228
		Edh at Noon	19	26	33	41	36	34	36	31	27	20	18	16
	RadAvg		344	679	1103	1490	1634	1629	1521	1268	956	565	295	224
(45) All-Sky Solar Radiation	RadStd		49	55	125	142	140	152	123	70	76	40	23	29

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (4 neighbors)		N/A	N/A	+2.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page